

Neural Network

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Read the dataset
data = pd.read_excel("/content/Student_Attendance_Dataset.xlsx")

# Display dataset
print("Dataset:")
print(data)

# Separate features and target
X = data.iloc[:, :-1].copy()
y = data.iloc[:, -1].copy()

# Encode categorical features
feature_encoders = {}

for column in X.columns:
    if X[column].dtype == 'object':
        encoder = LabelEncoder()
        X[column] = encoder.fit_transform(X[column])
        feature_encoders[column] = encoder

# Encode target
target_encoder = LabelEncoder()
y = target_encoder.fit_transform(y)

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Build Neural Network
```

```
model = Sequential()

# Input + Hidden Layer
model.add(Dense(8, input_dim=X_train.shape[1], activation='relu'))

# Second Hidden Layer
model.add(Dense(4, activation='relu'))

# Output Layer
model.add(Dense(1, activation='sigmoid'))

# Compile the model
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train the model
model.fit(X_train, y_train, epochs=100, batch_size=2, verbose=1)

# Predict
y_pred = model.predict(X_test)
y_pred = (y_pred > 0.5).astype(int)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nNeural Network Accuracy:", accuracy)

# Predict first test sample
prediction = model.predict(X_test.iloc[[0]])
prediction = (prediction > 0.5).astype(int)

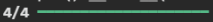
print("Predicted Class:",
      target_encoder.inverse_transform(prediction.flatten())[0])
```

OUTPUT:

```
target_encoder.inverse_transform(prediction.flatten())[0])

... Dataset:
  StudentID StudentName Department Attendance Assignment InternalMarks \
0      S001      Aarav      IT      Yes      Yes      82
1      S002      Diya      CSE      No      Yes      68
2      S003      Rahul      ECE      Yes      No      75
3      S004      Sneha      IT      Yes      Yes      91
4      S005      Kiran      MECH      No      No      55
5      S006      Priya      CSE      Yes      Yes      79
6      S007      Vikram      EEE      No      Yes      61
7      S008      Anu      IT      Yes      Yes      88
8      S009      Rohit      CIVIL      No      No      58
9      S010      Meena      ECE      Yes      Yes      84

  LabCompleted Eligible
0      Yes      Yes
1      No      No
2      Yes      Yes
3      Yes      Yes
4      No      No
5      Yes      Yes
6      No      No
7      Yes      Yes
8      No      No
9      Yes      Yes

Epoch 1/100
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim`
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
4/4  1s 6ms/step - accuracy: 0.8750 - loss: 0.5797
Epoch 2/100
```

Neural Network Accuracy: 0.5

1/1  0s 46ms/step